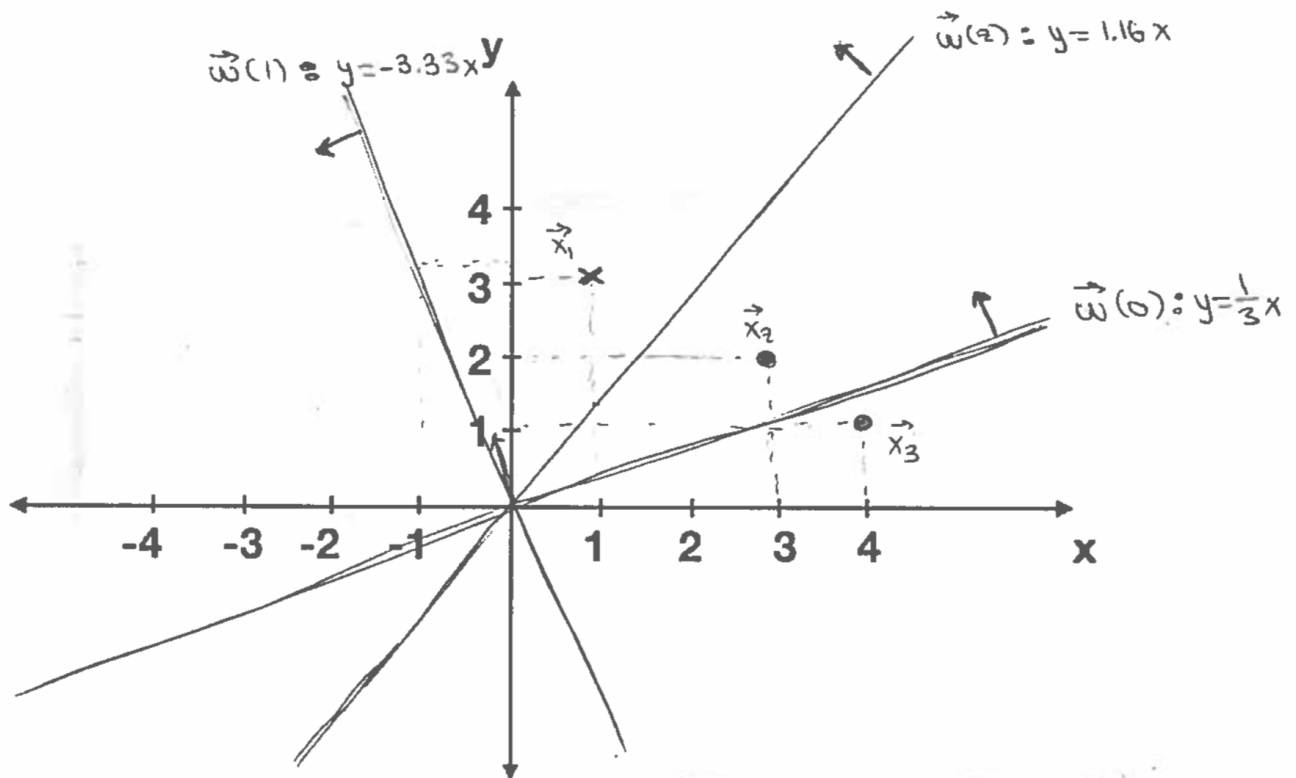


The goal of this problem is to run a linear perceptron algorithm. Assume that you have three training samples:

1. Sample $\mathbf{x}_1 = [1, 3]^T$ from Class 1 ($y_1 = 1$)
2. Sample $\mathbf{x}_2 = [3, 2]^T$ from Class 2 ($y_2 = -1$)
3. Sample $\mathbf{x}_3 = [4, 1]^T$ from Class 2 ($y_3 = -1$)

The linear perceptron is initialized with a line with corresponding weight $\mathbf{w}(0) = [-\frac{1}{3}, 1]^T$. In the following, for the sake of simplicity, you will assume that all lines of the perceptron intersect point $(0, 0)$, therefore **you do not have to include any intercept w_0 or x_0 in the following calculations.**



(1) Plot $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ in the given 2D space. Find and plot the line corresponding to weight $\mathbf{w}(0)$.

For $\mathbf{w}(0) = [-\frac{1}{3}, 1]^T$, we have $-\frac{1}{3}x + y = 0 \Rightarrow y = \frac{1}{3}x$, therefore the line is plot in the above figure. Note that the direction of the vector $\mathbf{w}_0$ on the line is the same as the direction of $\mathbf{w}(0)$ starting from $(0, 0)$.

(2) Using the rule $\text{sign}(\mathbf{w}(t)^T \mathbf{x}_n)$, please indicate in which class are samples $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ classified using the weight $\mathbf{w}(0)$. Which samples are not correctly classified based on this rule? **Note:** You have to compute the inner product $\mathbf{w}(0)^T \mathbf{x}_n$, $n = 1, 2, 3$, and see if it is greater or less than 0.

$\mathbf{w}(0)^T \mathbf{x}_1 = -\frac{1}{3} + 3 = \frac{8}{3} > 0$, therefore sample $\mathbf{x}_1$ is correctly classified using $\mathbf{w}(0)$.

$\mathbf{w}(0)^T \mathbf{x}_2 = -1 + 2 = 1 > 0$, therefore sample $\mathbf{x}_2$ is incorrectly classified using $\mathbf{w}(0)$.

$\mathbf{w}(0)^T \mathbf{x}_3 = -\frac{4}{3} + 1 = -\frac{1}{3} < 0$, therefore sample $\mathbf{x}_3$ is correctly classified using $\mathbf{w}(0)$.

(3) Using the weight update rule from the linear perceptron algorithm, please find the value of the new weight $\mathbf{w}(1)$. Find and plot the new line corresponding to weight $\mathbf{w}(1)$ in the 2D space.

Note: The update rule is $\mathbf{w}(\mathbf{t} + 1) = \mathbf{w}(\mathbf{t}) + y_s \mathbf{x}_s$, where $\mathbf{x}_s$ and $y_s \in \{-1, 1\}$ is the feature and class label of missclassified sample s .

$$\mathbf{w}(1) = \mathbf{w}(0) - \mathbf{x}_2 = \left[-\frac{1}{3}, 1\right]^T - [3, 2]^T = \left[-\frac{10}{3}, -1\right]^T.$$

Note the “-” sign in the above equation, which is because the misclassified sample from $\mathbf{w}(0)$, $\mathbf{x}_2$, belongs to Class 2 ($y_2 = -1$).

The corresponding line is $-\frac{10}{3}x - y = 0 \Rightarrow y = -\frac{10}{3}x \Rightarrow y = -3.33x$

(4) Using the rule $\text{sign}(\mathbf{w}(\mathbf{t})^T \mathbf{x}_n)$, please indicate in which class are samples $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ classified using the weight $\mathbf{w}(1)$. Which samples are not correctly classified based on this rule?

Note: You have to compute the inner product $\mathbf{w}(1)^T \mathbf{x}_n$, $n = 1, 2, 3$, and see if it is greater or less than 0.

$\mathbf{w}(1)^T \mathbf{x}_1 = -\frac{10}{3} - 3 = -\frac{19}{3} < 0$, therefore sample $\mathbf{x}_1$ is incorrectly classified using $\mathbf{w}(1)$.

$\mathbf{w}(1)^T \mathbf{x}_2 = -10 - 2 = -12 < 0$, therefore sample $\mathbf{x}_2$ is correctly classified using $\mathbf{w}(1)$.

$\mathbf{w}(1)^T \mathbf{x}_3 = -\frac{40}{3} - 1 = -\frac{43}{3} < 0$, therefore sample $\mathbf{x}_3$ is correctly classified using $\mathbf{w}(1)$.

(5) Using the weight update rule from the linear perceptron algorithm, please find the value of the new weight $\mathbf{w}(2)$. Find and plot the new line corresponding to weight $\mathbf{w}(2)$ in the 2D space. How many samples are correctly classified now?

Note: The update rule is $\mathbf{w}(\mathbf{t} + 1) = \mathbf{w}(\mathbf{t}) + y_s \mathbf{x}_s$, where $\mathbf{x}_s$ and $y_s \in \{-1, 1\}$ is the feature and class label of missclassified sample s .

$$\mathbf{w}(2) = \mathbf{w}(1) + \mathbf{x}_1 = \left[-\frac{10}{3}, -1\right]^T + [1, 3]^T = \left[-\frac{7}{3}, 2\right]^T.$$

Note the “+” sign in the above equation, which is because the misclassified sample from $\mathbf{w}(1)$, $\mathbf{x}_1$, belongs to Class 1 ($y_1 = 1$).

The corresponding line is $-\frac{7}{3}x + 2y = 0 \Rightarrow y = \frac{7}{6}x \Rightarrow y = 1.16x$

$\mathbf{w}(2)^T \mathbf{x}_1 = -\frac{7}{3} + 6 = \frac{11}{3} > 0$, therefore sample $\mathbf{x}_1$ is correctly classified using $\mathbf{w}(2)$.

$\mathbf{w}(2)^T \mathbf{x}_2 = -7 + 4 = -3 < 0$, therefore sample $\mathbf{x}_2$ is correctly classified using $\mathbf{w}(2)$.

$\mathbf{w}(2)^T \mathbf{x}_3 = -\frac{28}{3} + 2 = -\frac{22}{3} < 0$, therefore sample $\mathbf{x}_3$ is correctly classified using $\mathbf{w}(2)$.

All samples are classified correctly based on $\mathbf{w}(2)$.